Name: J Mukhesh H.NO:2303A51813 BATCH:26

| SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE | | DEPARTMENT OF COMPUTER SCIENCE ENGINEERING | |
|---|---|---|---|
| **Program Name:** B. Tech | | **Assignment Type: Lab** | **Academic Year:**2025-2026 |
| **Course Coordinator Name** | Dr. Rishabh Mittal | | |
| **Instructor(s) Name** | Mr. S Naresh Kumar<br>Ms. B. Swathi<br>Dr. Sasanko Shekhar Gantayat<br>Mr. Md Sallauddin<br>Dr. Mathivanan<br>Mr. Y Srikanth<br>Ms. N Shilpa<br>Dr. Rishabh Mittal (Coordinator)<br>Dr. R. Prashant Kumar<br>Mr. Ankushavali MD<br>Mr. B Viswanath<br>Ms. Sujitha Reddy<br>Ms. A. Anitha<br>Ms. M.Madhuri<br>Ms. Katherashala Swetha<br>Ms. Velpula sumalatha<br>Mr. Bingi Raju | | |
| **CourseCode** | 23CS002PC304 | **Course Title** | AI Assisted Coding |
| **Year/Sem** | III/II | **Regulation** | R23 |
| **Date and Day of Assignment** | **Week10 – Friday** | **Time(s)** | 23CSBTB01 To 23CSBTB52 |
| **Duration** | 2 Hours | **Applicable to Batches** | All batches |
| **Assignment Number:20.5**(Present assignment number)/**24**(Total number of assignments) | | | |

| Q.No. | Question | ***Expected Time to complete*** |
|---|---|---|
| 1 | **Lab 20 – Security Testing: Identifying Vulnerabilities in AI-Generated Code** | Week10 - Friday |

**Lab Objectives:**

- Understand how to test AI-generated code for common security vulnerabilities.
- Learn to apply secure coding principles while analyzing AI outputs.
- Practice detecting risks such as SQL injection, XSS, hardcoded credentials, and weak encryption.
- Enhance code reliability and safety by using AI for secure refactoring.

**Task 1 – Password Storage Vulnerability Check**

**Task:**

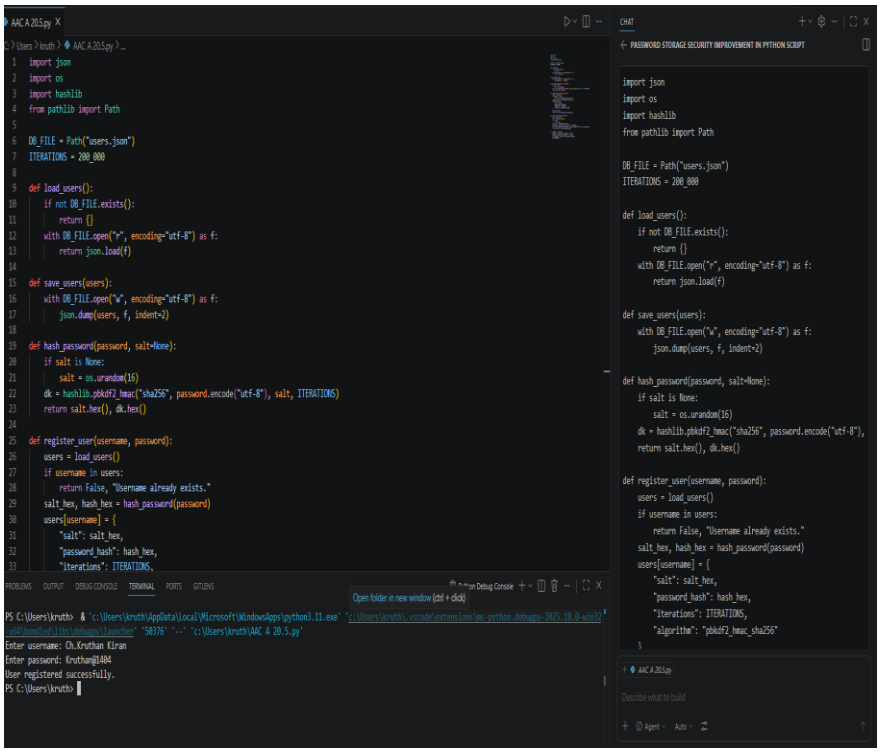
Analyze an AI-generated signup script for insecure password storage.

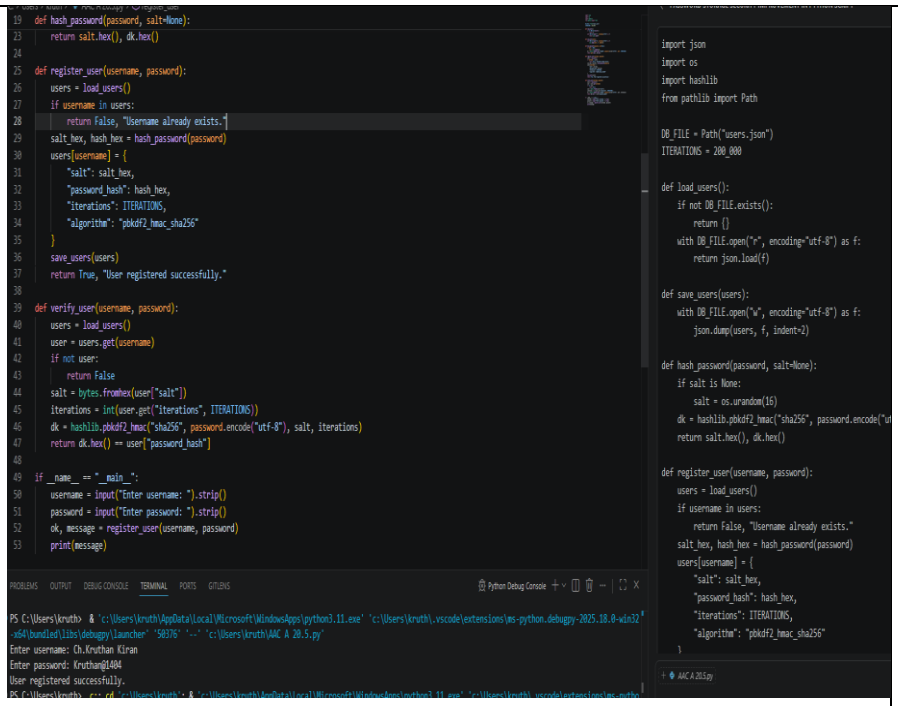
**Instructions:**

- Prompt AI to generate a Python user registration program.
- Check whether passwords are stored in plain text.
- Improve the script using hashing (bcrypt or hashlib).

**Expected Output:**

- A secure registration script with hashed password storage.





```
def hash_password(password, salt=None):
    return salt.hex(), dk.hex()

def register_user(username, password):
    users = load_users()
    if username in users:
        return False, "Username already exists."
    salt_hex, hash_hex = hash_password(password)
    users[username] = {
        "salt": salt_hex,
        "password_hash": hash_hex,
        "iterations": ITERATIONS,
        "algorithm": "pbkdf2_hmac_sha256"
    }
    save_users(users)
    return True, "User registered successfully."

def verify_user(username, password):
    users = load_users()
    user = users.get(username)
    if not user:
        return False
    salt = bytes.fromhex(user["salt"])
    iterations = int(user.get("iterations", ITERATIONS))
    dk = hashlib.pbkdf2_hmac("sha256", password.encode("utf-8"), salt, iterations)
    return dk.hex() == user["password_hash"]

if __name__ == "__main__":
    username = input("Enter username: ").strip()
    password = input("Enter password: ").strip()
    ok, message = register_user(username, password)
    print(message)
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS
Python Debug Console

PS C:\Users\kruth> & "c:\Users\kruth\AppData\Local\Microsoft\WindowsApps\python3.11.exe" "c:\Users\kruth\.vscode\extensions\ms-python.debugpy-2025.18.0-win32
-x64\bundled\libs\debugpy\launcher" "58376" "--" "c:\Users\kruth\AAC A 20.5.py"
Enter username: Ch.Kruthan Kiran
Enter password: Kruthan@1404
User registered successfully.
```

```
import json
import os
import hashlib
from pathlib import Path

DB_FILE = Path("users.json")
ITERATIONS = 200_000

def load_users():
    if not DB_FILE.exists():
        return {}
    with DB_FILE.open("r", encoding="utf-8") as f:
        return json.load(f)

def save_users(users):
    with DB_FILE.open("w", encoding="utf-8") as f:
        json.dump(users, f, indent=2)

def hash_password(password, salt=None):
    if salt is None:
        salt = os.urandom(16)
    dk = hashlib.pbkdf2_hmac("sha256", password.encode("ut
    return salt.hex(), dk.hex()

def register_user(username, password):
    users = load_users()
    if username in users:
        return False, "Username already exists."
    salt_hex, hash_hex = hash_password(password)
    users[username] = {
        "salt": salt_hex,
        "password_hash": hash_hex,
        "iterations": ITERATIONS,
        "algorithm": "pbkdf2_hmac_sha256"
    }
```

AAC A 20.5.py

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**Task 2 – Hardcoded Secrets Detection**

**Task:**

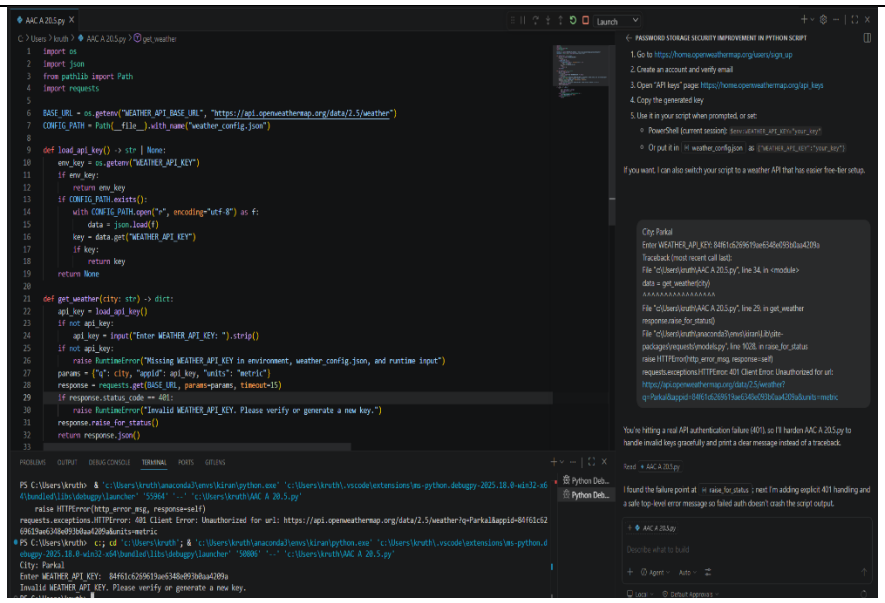
Evaluate AI-generated code for hardcoded API keys or credentials.

**Instructions:**

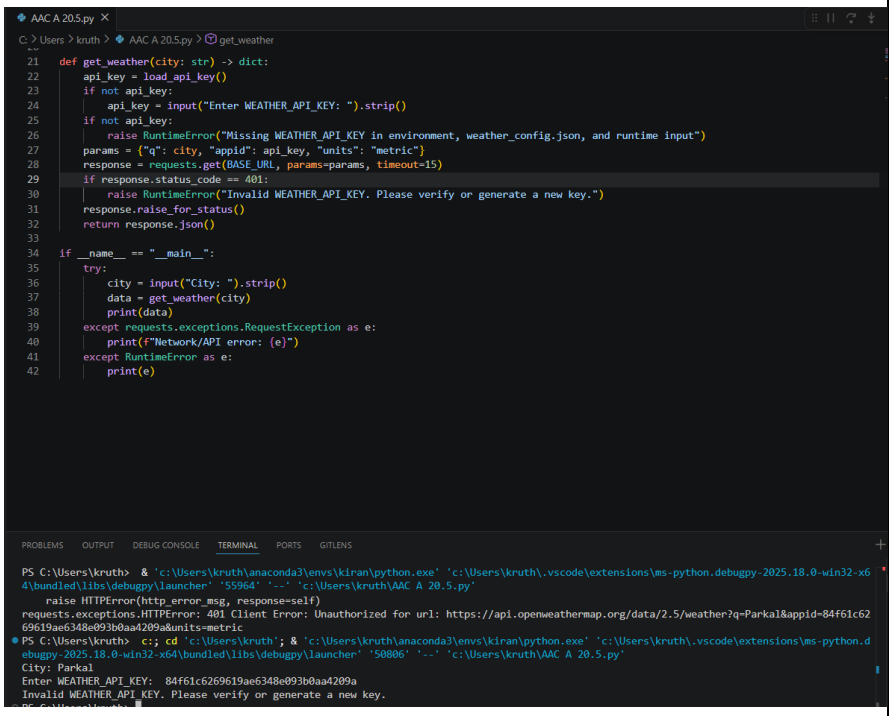
- Ask AI to generate a script that connects to an external API.
- Identify if API keys/passwords are written directly in code.
- Refactor using environment variables or config files.

**Expected Output:**

- A secure script with secrets managed properly.



```
AAC A 20.5.py
C: > Users > kruth > AAC A 20.5.py > get_weather
import os
import json
from pathlib import Path
import requests

BASE_URL = os.getenv("WEATHER_API_BASE_URL", "https://api.openweathermap.org/data/2.5/weather")
CONFIG_PATH = Path(__file__).with_name("weather_config.json")

def load_api_key() -> str | None:
    env_key = os.getenv("WEATHER_API_KEY")
    if env_key:
        return env_key
    if CONFIG_PATH.exists():
        with CONFIG_PATH.open("r", encoding="utf-8") as f:
            data = json.load(f)
        key = data.get("WEATHER_API_KEY")
        if key:
            return key
    return None

def get_weather(city: str) -> dict:
    api_key = load_api_key()
    if not api_key:
        api_key = input("Enter WEATHER_API_KEY: ").strip()
    if not api_key:
        raise RuntimeError("Missing WEATHER_API_KEY in environment, weather_config.json, and runtime input")
    params = {"q": city, "appid": api_key, "units": "metric"}
    response = requests.get(BASE_URL, params=params, timeout=15)
    if response.status_code == 401:
        raise RuntimeError("Invalid WEATHER_API_KEY. Please verify or generate a new key.")
    response.raise_for_status()
    return response.json()

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS

PS C:\Users\kruth>  & 'c:\Users\kruth\anaconda3\envs\kiran\python.exe' 'c:\Users\kruth\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '55964' '--' 'c:\Users\kruth\AAC A 20.5.py'
    raise HTTPError(http_error_msg, response=self)
requests.exceptions.HTTPError: 401 Client Error: Unauthorized for url: https://api.openweathermap.org/data/2.5/weather?q=Parkal&appid=84f61c6269619ae6348e093b0aa4209a&units=metric
PS C:\Users\kruth>  c:; cd 'c:\Users\kruth'; & 'c:\Users\kruth\anaconda3\envs\kiran\python.exe' 'c:\Users\kruth\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '50806' '--' 'c:\Users\kruth\AAC A 20.5.py'
City: Parkal
Enter WEATHER_API_KEY:  84f61c6269619ae6348e093b0aa4209a
Invalid WEATHER_API_KEY. Please verify or generate a new key.
```



```
AAC A 20.5.py
C: > Users > kruth > AAC A 20.5.py > get_weather
def get_weather(city: str) -> dict:
    api_key = load_api_key()
    if not api_key:
        api_key = input("Enter WEATHER_API_KEY: ").strip()
    if not api_key:
        raise RuntimeError("Missing WEATHER_API_KEY in environment, weather_config.json, and runtime input")
    params = {"q": city, "appid": api_key, "units": "metric"}
    response = requests.get(BASE_URL, params=params, timeout=15)
    if response.status_code == 401:
        raise RuntimeError("Invalid WEATHER_API_KEY. Please verify or generate a new key.")
    response.raise_for_status()
    return response.json()

if __name__ == "__main__":
    try:
        city = input("City: ").strip()
        data = get_weather(city)
        print(data)
    except requests.exceptions.RequestException as e:
        print(f"Network/API error: {e}")
    except RuntimeError as e:
        print(e)

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  GITLENS

PS C:\Users\kruth>  & 'c:\Users\kruth\anaconda3\envs\kiran\python.exe' 'c:\Users\kruth\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '55964' '--' 'c:\Users\kruth\AAC A 20.5.py'
    raise HTTPError(http_error_msg, response=self)
requests.exceptions.HTTPError: 401 Client Error: Unauthorized for url: https://api.openweathermap.org/data/2.5/weather?q=Parkal&appid=84f61c6269619ae6348e093b0aa4209a&units=metric
PS C:\Users\kruth>  c:; cd 'c:\Users\kruth'; & 'c:\Users\kruth\anaconda3\envs\kiran\python.exe' 'c:\Users\kruth\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '50806' '--' 'c:\Users\kruth\AAC A 20.5.py'
City: Parkal
Enter WEATHER_API_KEY:  84f61c6269619ae6348e093b0aa4209a
Invalid WEATHER_API_KEY. Please verify or generate a new key.
```

## Task 3 – File Upload Security Validation

**Task:**

Test an AI-generated file upload program for unsafe file handling.

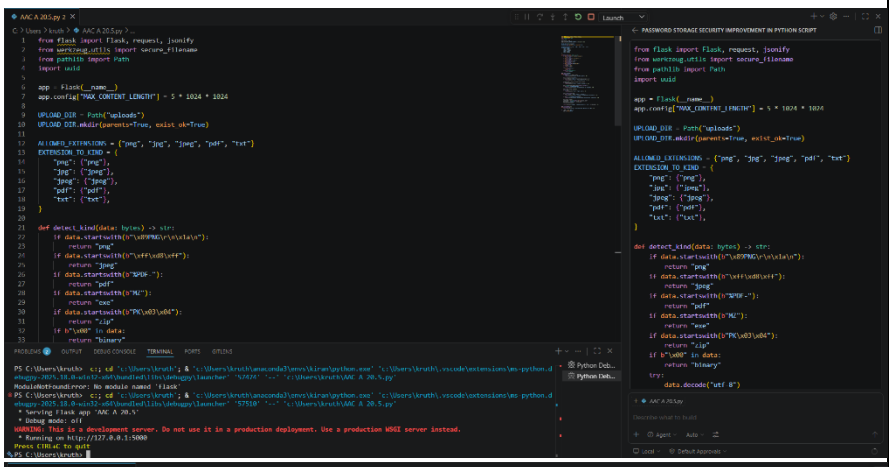
**Instructions:**

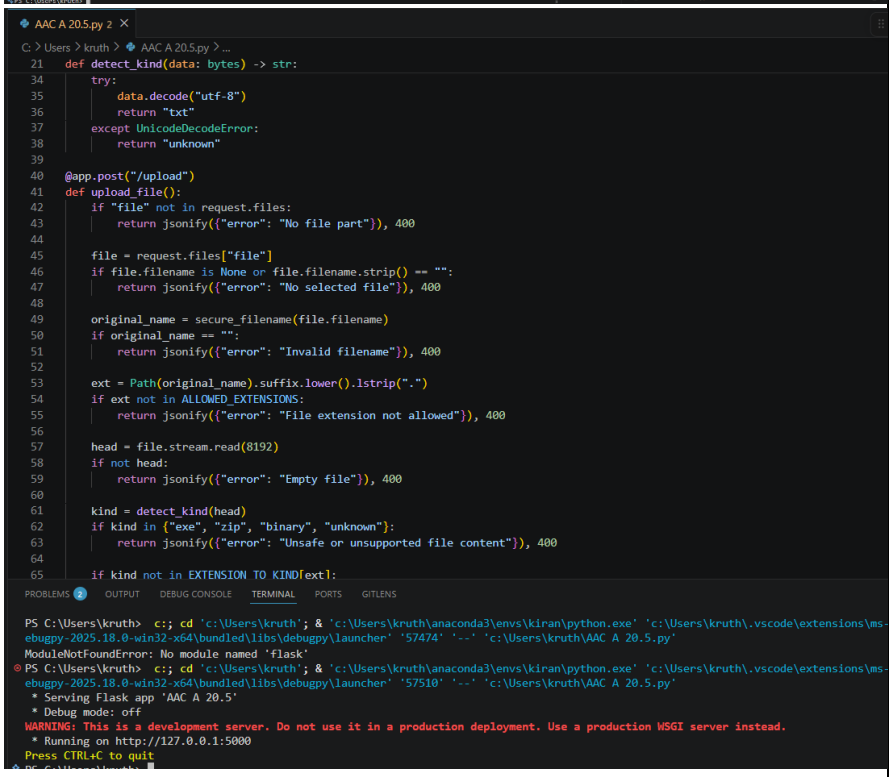
- Prompt AI to generate a Flask file upload example.

- Check for vulnerabilities like uploading executable files.
- Add validation for file type, size, and secure filename handling.

**Expected Output:**

- A secure file upload implementation.





```python
def upload_file():
    if original_name == "":
        return jsonify({"error": "Invalid filename"}), 400

    ext = Path(original_name).suffix.lower().lstrip(".")
    if ext not in ALLOWED_EXTENSIONS:
        return jsonify({"error": "File extension not allowed"}), 400

    head = file.stream.read(8192)
    if not head:
        return jsonify({"error": "Empty file"}), 400

    kind = detect_kind(head)
    if kind in {"exe", "zip", "binary", "unknown"}:
        return jsonify({"error": "Unsafe or unsupported file content"}), 400

    if kind not in EXTENSION_TO_KIND[ext]:
        return jsonify({"error": "File content does not match extension"}), 400

    file.stream.seek(0)
    stored_name = f"{uuid.uuid4().hex}_{original_name}"
    destination = UPLOAD_DIR / stored_name
    file.save(destination)

    return jsonify({"message": "Upload successful", "file": stored_name}), 201

@app.errorhandler(413)
def file_too_large(_):
    return jsonify({"error": "File too large. Max size is 5 MB"}), 413

if __name__ == "__main__":
    app.run(debug=False)
```

```
PS C:\Users\kruth>  c:; cd 'c:\Users\kruth'; & 'c:\Users\kruth\anaconda3\envs\kiran\python.exe' 'c:\Users\kruth\.vscode\ex
ebugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '57474' '--' 'c:\Users\kruth\AAC A 20.5.py'
ModuleNotFoundError: No module named 'flask'
PS C:\Users\kruth>  c:; cd 'c:\Users\kruth'; & 'c:\Users\kruth\anaconda3\envs\kiran\python.exe' 'c:\Users\kruth\.vscode\ex
ebugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '57510' '--' 'c:\Users\kruth\AAC A 20.5.py'
 * Serving Flask app 'AAC A 20.5'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
```

## Task 4 – Command Injection Prevention

**Task:**

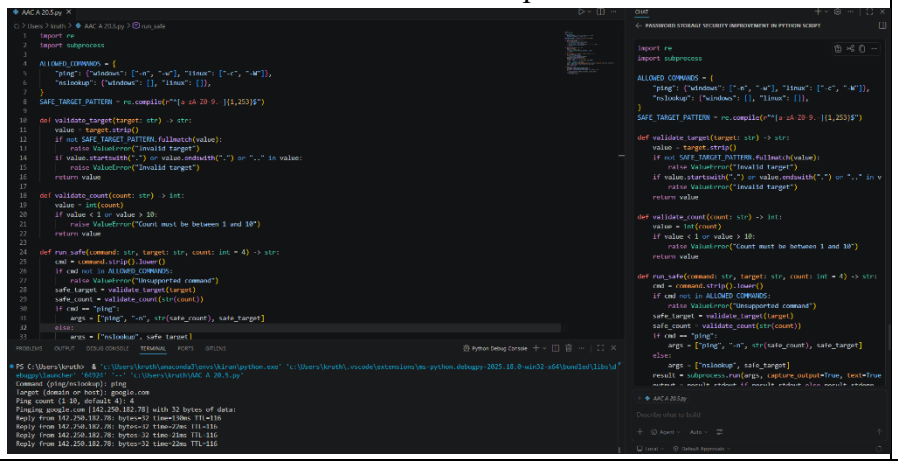
Analyze an AI-generated Python script that executes system commands.

**Instructions:**

- Ask AI to generate code using os.system() or subprocess.
- Identify command injection risks with user input.
- Fix using safe subprocess calls and input validation.

**Expected Output:**

- A secure command execution script.



AAC A 20.5.py

C: > Users > kruth > AAC A 20.5.py > run_safe

```python
def run_safe(command: str, target: str, count: int = 4) -> str:
    cmd = command.strip().lower()
    if cmd not in ALLOWED_COMMANDS:
        raise ValueError("Unsupported command")
    safe_target = validate_target(target)
    safe_count = validate_count(str(count))
    if cmd == "ping":
        args = ["ping", "-n", str(safe_count), safe_target]
    else:
        args = ["nslookup", safe_target]
    result = subprocess.run(args, capture_output=True, text=True, timeout=15, shell=False, check=False)
    output = result.stdout if result.stdout else result.stderr
    return output.strip()

if __name__ == "__main__":
    user_command = input("Command (ping/nslookup): ")
    user_target = input("Target (domain or host): ")
    user_count = input("Ping count (1-10, default 4): ").strip() or "4"
    try:
        print(run_safe(user_command, user_target, int(user_count)))
    except Exception as e:
        print(f"Error: {e}")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

```
PS C:\Users\kruth>  & 'c:\Users\kruth\anaconda3\envs\kiran\python.exe' 'c:\Users\kruth\.vscode\extensions\ms-python.
ebugpy\launcher' '64924' '--' 'c:\Users\kruth\AAC A 20.5.py'
Reply from 142.250.182.78: bytes=32 time=22ms TTL=116
Reply from 142.250.182.78: bytes=32 time=21ms TTL=116
Reply from 142.250.182.78: bytes=32 time=22ms TTL=116

Ping statistics for 142.250.182.78:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 21ms, Maximum = 130ms, Average = 48ms
PS C:\Users\kruth>
```

## Task 5 – Secure Input Handling in Web Forms

**Task:**

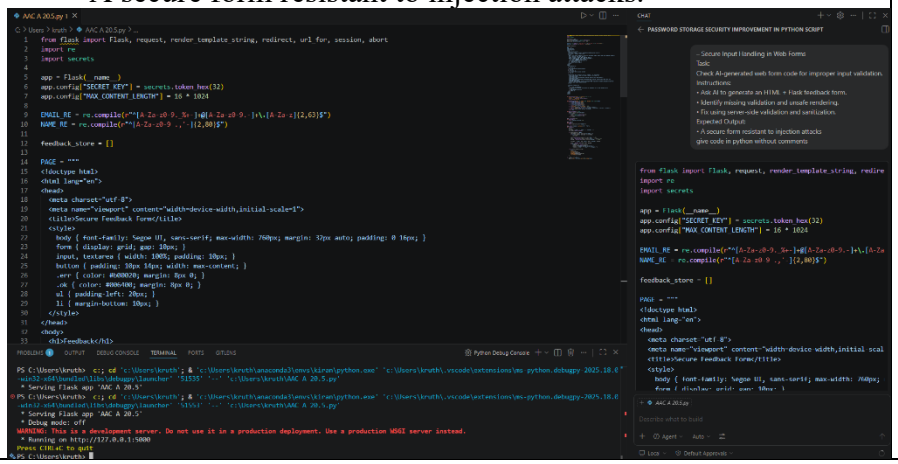
Check AI-generated web form code for improper input validation.

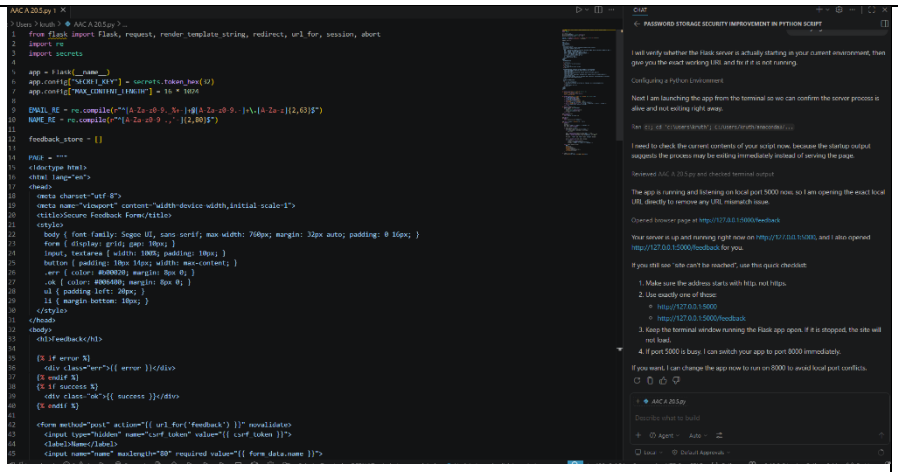
**Instructions:**

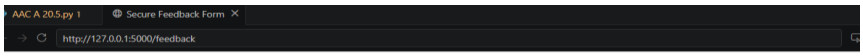
- Ask AI to generate an HTML + Flask feedback form.
- Identify missing validation and unsafe rendering.
- Fix using server-side validation and sanitization.

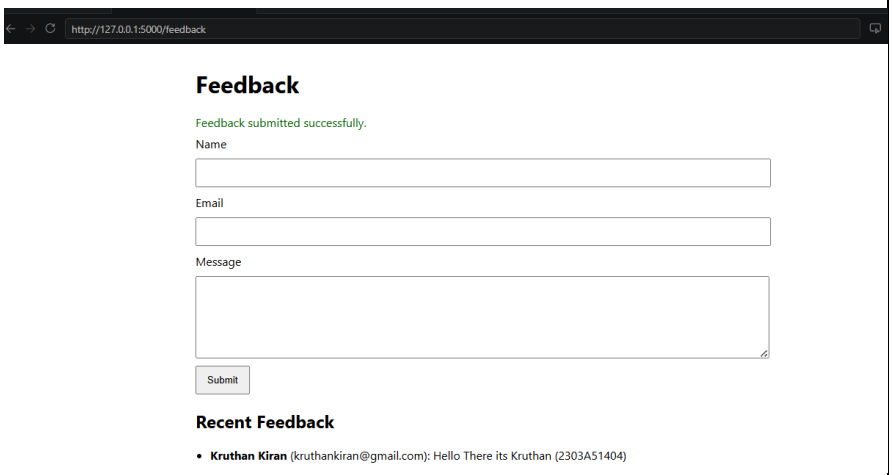
**Expected Output:**

- A secure form resistant to injection attacks.








Note: Report should be submitted as a word document for all tasks in a single document with prompts, comments & code explanation, and output and if required, screenshots.